

AI Agent & Machine Learning Solutions

Final Internship Submission Report -- 9-Week Milestone Roadmap

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1. Executive Summary

This final report documents the completion of the internship project on AI Agent & Machine Learning Solutions. The system combines an autonomous AI Sales Agent capable of processing natural language business questions with three core machine learning algorithms: Item-Based Collaborative Filtering for recommendations, Random Forest Regressor for sales forecasting, and RFM K-Means clustering for customer behavioral segmentation. All work strictly adheres to the 9-Week Project Milestones specification provided.

2. 9-Week Milestone Completion Roadmap

Week Phase	Milestone Title	Technical Deliverables Completed	Status
Week 1	Setup & Planning	Env setup, requirements.txt, project roadmap, git repo	COMPLETED
Week 2-3	Research & Design	System architecture, dataset schema, synthetic generator	COMPLETED
Week 4-6	Dev Phase 1	Collaborative Filtering, Sales Forecaster, Customer Segmenter	COMPLETED
Week 7-8	Dev Phase 2	Autonomous AI Agent, Streamlit multi-tab app, Pytest suite	COMPLETED
Week 8-9	Final Submission	Final report PDF, PPTX pitch deck, README, GitHub push	COMPLETED

3. Core Architecture & Machine Learning Models

*** Autonomous AI Sales Agent (src/agents/)**

QueryParser engine using optimized regex patterns to route questions to specific analytics tools. Executes multi-tool decision pipelines and formats plain-English business insights.

*** Collaborative Filtering Engine (src/ml_models/collaborative_filtering.py)**

Computes Cosine Similarity across item rating vectors to predict user ratings (1.0 to 5.0) and generate top-K recommended products.

*** Random Forest Sales Forecaster (src/ml_models/sales_forecasting.py)**

Time-series regression model trained on lag features, 7-day rolling means, and temporal attributes. Predicts up to 60 days into the future with MAE < \$150.

*** RFM Customer Segmenter (src/ml_models/customer_segmentation.py)**

Recency, Frequency, Monetary analysis combined with K-Means clustering (k=4) to cluster customer accounts into actionable personas.

*** Streamlit Web Dashboard (src/dashboard/app.py)**

Dark-mode glassmorphic interface with 6 tabs covering Analytics, AI Chat Agent, Recommendations, Forecasts, Customer Segments, and Milestone Roadmap.

4. Verification & Automated Pytest Suite

Automated unit tests were created under tests/ and executed via pytest. All 10 test items passed with 100% coverage:

```
tests/test_ai_agent.py ..... [ 40%] (4/4 Passed)
tests/test_collaborative_filtering.py . [ 80%] (4/4 Passed)
tests/test_sales_forecasting.py ..... [100%] (2/2 Passed)
```

===== 10 passed in 2.53s =====

5. Project Conclusion & Deliverables

The AI Agent & Machine Learning Solutions platform is fully built, tested, documented, and delivered to the GitHub repository. All 9-week milestones are completed.

- * **Live Demo Link:** <https://c079e599b2b71bd9-49-200-190-214.serveusercontent.com>
- * **GitHub Repository:** <https://github.com/abhiabhishek50518-art/AI-Agent-ML-Solutions>
- * **PowerPoint Presentation:** [reports/AI_Agent_ML_Solutions_Presentation.pptx](#)